Gatekeeper

System Description

Abstract
This document describes the Gatekeeper support system of the Eclipse Arrowhead 5.2.2 framework, as implemented in `ah5-Gatekeeper-java-spring` with an independent local-cloud deployment definition in its own folder.

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1 Overview

This document describes the Gatekeeper system. The Gatekeeper connects the local cloud to its neighbour clouds through the inter-cloud MQTT relay: global service discovery of authorized foreign providers and inter-cloud negotiation of GateTunnel sessions.

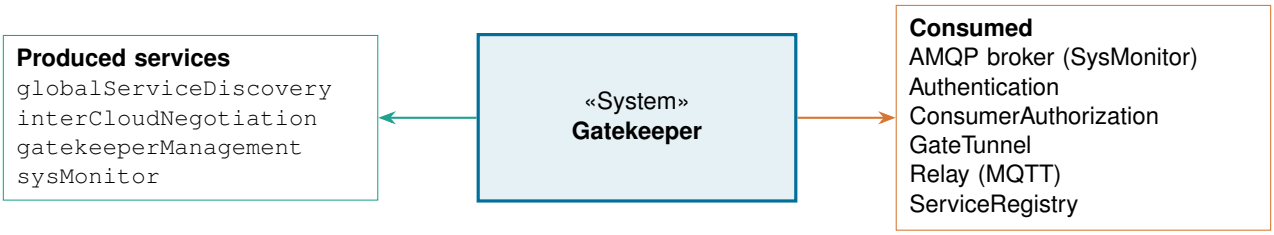


Figure 1: SysML-style block view of Gatekeeper with its produced and consumed services.

The rest of this document is organized as follows. In Section 1.1, we reference major prior art capabilities of the system. In Section 1.2, we describe the intended usage of the system. In Section 1.3, we describe fundamental properties provided by the system. In Section 1.4, we describe delimitations of capabilities of the system. In Section 2, we describe the abstract service functions consumed or produced by the system. In Section 3, we describe the security capabilities of the system.

1.1 Significant Prior Art

Successor of the Arrowhead 4.6 Gatekeeper; native Java port of the former Python implementation (Python sources kept for rollback).

1.2 How This System Is Meant to Be Used

DynamicServiceOrchestration calls `globalServiceDiscovery.query` and `interCloudNegotiation.negotiate` during inter-cloud orchestration; the result points the consumer at the local GateTunnel.

1.3 System functionalities and properties

1.3.1 Functional properties of the system

- GSD over the relay with peer-identity check.
- Provider-cloud authorization with ConsumerAuthorization `check-policies`.
- ICN with GateTunnel provider and consumer sessions.
- Configuration of clouds and relays.

1.3.2 Configuration of system properties

The system is configured with environment variables. The variables that shape its behaviour are:

Property	Default	Meaning
`PORT`	8449	HTTPS port.
`RELAY_HOST` / `RELAY_PORT`	arrowhead-relay / 8883	Inter-cloud relay (MQTT over TLS).
`ORGANIZATION` / `FEDERATION_PEER_ORGANIZATIONS`	Company / (empty)	Own and peer organizations.
`FEDERATION_PROFILE`	legacy-mtls	`legacy-mtls` or `ah-gt-1`.
`SERVICE_REGISTRY_URL`	`https://serviceregistry:8443`	Registry.

Secrets (database, keystore and identity passwords, secret keys) have no usable defaults; the standalone deployment reads them from per-deployment Compose file secrets under `/run/secrets`.

1.3.3 Data stored by the system

Neighbour clouds and relays (SQLite).

1.3.4 Non functional properties

Relay requests have timeouts; duplicate QoS 1 requests reuse cached responses.

1.3.5 Stateful or stateless

Stateful configuration; sessions are held by GateTunnel.

1.4 Important Delimitations

`legacy-mtls` is not end-to-end encrypted: the relay can see the payload. AH-GT-1 peers are refused, not downgraded.

2 Services

This section describes the produced and consumed services. Each produced service is specified by an SD document and one IDD document per interface.

2.1 Produced services

Service	Version	Interfaces	SD	IDD
`globalServiceDiscovery`	5.2.2	HTTP(S)	[1]	[2]
`interCloudNegotiation`	5.2.2	HTTP(S)	[3]	[4]
`gatekeeperManagement`	5.2.2	HTTP(S)	[5]	[6]
`sysMonitor`	1.0.0	AMQP(S)	[7]	[8]

All services use `generic_https` with `CERT_AUTH`; `sysMonitor` uses `generic_amqps`. The container also answers `GET /health` on a loopback-only plain-HTTP port for the container health check.

2.2 Consumed services

Provider	Service	Operations	Use	Condition
ServiceRegistry	serviceDiscovery	register, lookup	Registration; local providers.	always
Authentication	identity	identity-login, identity-verify	Own token; caller verification.	always
Consumer Authorization	authorization Management	check-policies	Authorization of foreign consumers.	provider cloud
GateTunnel	gateTunnel Connection	open-provider-session, open-consumer-session	Relay sessions.	negotiation
Relay (MQTT)	inter-cloud messages	gsd-request, icn-request	Neighbour Gatekeepers.	always
AMQP broker (SysMonitor)	sysMonitor (publisher)	echo, power, status	Own observations.	always

Only Authentication's `identity` and the orchestrator are looked up in the ServiceRegistry; other services are reached through DynamicServiceOrchestration (results cached), and every such call needs a Consumer-Authorization grant.

3 Security

The standalone Java system requires HTTPS with mutual TLS and X.509 certificates following the Arrowhead system certificate profile (`<System>.<Cloud>.<Organization>.arrowhead.eu`, DNQ `sy`). There is no insecure runtime mode. Identity tokens are verified through Authentication and bound to the caller certificate. A separate loopback health endpoint is infrastructure only. Provision credentials and a valid CRL using the project-local environment template and Compose definition.

3.1 Security model

- **Protocols:** HTTPS (TLS 1.3 and 1.2) with mandatory client certificates; MQTT over TLS; AMQP over TLS for `sysMonitor`.
- **Data protection:** transport encryption by TLS; SQLite storage is local to this system's private volume. Passwords are read from Compose file secrets; certificate and private-key files are mounted individually.
- **System authentication:** mutual TLS with a system or operator certificate of the cloud and an identity token of the same system, verified with Authentication; declared identities are not accepted. Client certificates are checked against the CRL.
- **Authorization of produced services:** per operation by caller lists in the implementation and, where stated, by ConsumerAuthorization.
- Service operations only for DynamicServiceOrchestration, SimpleStoreServiceOrchestration, LocalCloudManager, Gatekeeper and Sysop; management only LocalCloudManager and Sysop. Only orchestrators may act for another consumer.

For the Arrowhead certificate profile see github.com/eclipse-arrowhead/documentation.

4 References

[1] J. Delsing, "globalServiceDiscovery – Service Description (Gatekeeper)," Eclipse Arrowhead, SD 5.2.2, 2026, gatekeeper-v5.2.2-globalServiceDiscovery-SD.pdf.

[2] ——, "globalServiceDiscovery HTTP/TLS/JSON – Interface Design Description (Gatekeeper)," Eclipse Arrowhead, IDD 5.2.2, 2026, gatekeeper-v5.2.2-globalServiceDiscovery-HTTP-IDD.pdf.

[3] ——, "interCloudNegotiation – Service Description (Gatekeeper)," Eclipse Arrowhead, SD 5.2.2, 2026, gatekeeper-v5.2.2-interCloudNegotiation-SD.pdf.

[4] ——, "interCloudNegotiation HTTP/TLS/JSON – Interface Design Description (Gatekeeper)," Eclipse Arrowhead, IDD 5.2.2, 2026, gatekeeper-v5.2.2-interCloudNegotiation-HTTP-IDD.pdf.

[5] ——, "gatekeeperManagement – Service Description (Gatekeeper)," Eclipse Arrowhead, SD 5.2.2, 2026, gatekeeper-v5.2.2-gatekeeperManagement-SD.pdf.

[6] ——, "gatekeeperManagement HTTP/TLS/JSON – Interface Design Description (Gatekeeper)," Eclipse Arrowhead, IDD 5.2.2, 2026, gatekeeper-v5.2.2-gatekeeperManagement-HTTP-IDD.pdf.

[7] ——, "sysMonitor – Service Description (Gatekeeper)," Eclipse Arrowhead, SD 5.2.2, 2026, gatekeeper-v5.2.2-sysMonitor-SD.pdf.

[8] ——, "sysMonitor AMQP/TLS/JSON – Interface Design Description (Gatekeeper)," Eclipse Arrowhead, IDD 5.2.2, 2026, gatekeeper-v5.2.2-sysMonitor-AMQP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-25	5.2.2	Independent Java build, private helpers and state, per-system deployment and configuration.	Codex (draft)
3	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	